ABEBE B. ADEGE, Ph.D.

Data Science | Remote Sensing | Computer Vision

Machine Learning | Multimodal Data Fusion | Spatiotemporal Modeling | Intelligent Sensing

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PROFESSIONAL PROFILE

Computer scientist, data scientist, and AI/ML researcher with 10+ years of experience developing data-driven methods and intelligent systems, with primary expertise in data science, remote sensing, and computer vision. My research integrates satellite and UAV imagery, geospatial observations, sensor and IoT measurements, weather and other spatiotemporal data to support detection, monitoring, prediction, forecasting, and decision-making. I develop machine-learning, deep-learning, statistical, and multimodal data-fusion approaches for image analysis, geospatial analytics, time-series modeling, localization, object detection and tracking, and robust inference from heterogeneous observations. My work is method-centered and application-driven, with experience across environmental and agricultural systems, climate-related analysis, public safety, connected systems, and health. I also have a strong record of peer-reviewed publications, externally funded interdisciplinary research, student supervision, research leadership, and high-performance/GPU computing.

RESEARCH FOCUS & CORE EXPERTISE

- Data Science & Predictive Analytics: Machine learning, statistical modeling, supervised and unsupervised learning, ensemble methods, feature engineering, predictive analytics, time-series forecasting, spatiotemporal modeling, model validation, and large-scale heterogeneous data analysis.
- Remote Sensing & GeoAI: Satellite and UAV imagery, geospatial data integration, spatial analysis, image mosaicking and mapping, Google Earth Engine, QGIS, ArcGIS, Pix4D, and remote-sensing-based monitoring and prediction.
- Computer Vision & Visual Intelligence: Deep learning for object detection, image classification, segmentation, tracking, feature extraction, transfer learning, multimodal image analysis, and vision-based sensing from UAV and ground platforms.
- Intelligent Sensing & Multimodal Data Fusion: IoT and environmental sensing, heterogeneous sensor integration, weather and soil sensing, real-time data acquisition, multimodal fusion, sensor-data quality assessment, and robust inference from noisy observations.
- Computational AI & Application Domains: Deep learning, RNN/GRU/BiLSTM/CNN/Transformer methods, HPC/GPU computing, scalable workflows, and interdisciplinary applications in environmental and agricultural systems, climate, disaster and public safety, smart/connected systems, and health.

EDUCATION

Ph.D., Electrical Engineering and Computer Science, National Taipei University of Technology, Taipei, Taiwan, 2020.

Dissertation: Deep Machine Learning Approaches for Position-Aware Services in Internet of Things Networks.

M.Sc., Information Science, Addis Ababa University, Addis Ababa, Ethiopia, 2011.

Thesis: Rule-based Information Retrieval for Ge'ez Text.

B.Sc., Computer Science, Addis Ababa University, Addis Ababa, Ethiopia, 2007.

Project: Design of an Expert System for Diabetic Disease Diagnosis in an AI System.

RESEARCH & ACADEMIC EXPERIENCE

University of Florida, Institute of Food and Agricultural Sciences (UF/IFAS), Tropical Research and Education Center (TREC) - Postdoctoral Research Associate | *USA*, Jun 2024 - Present

- Develop data-science and multimodal AI workflows that integrate UAV and satellite imagery, geospatial observations, environmental sensors, weather data, and field measurements for monitoring, prediction, forecasting, and decision-support applications.
- Conduct UAV-based data collection and develop computer-vision and deep-learning pipelines for image stitching, mapping, ground-control-point detection, field-boundary extraction, object and condition monitoring, water-stress analysis, disease identification, and environmental assessment.
- Manage and analyze high-resolution satellite, UAV, and geospatial datasets using Google Earth Engine, QGIS, ArcGIS, Pix4D, and cloud-based computational workflows for spatial analysis and remote-sensing applications.
- Develop scalable spatiotemporal machine-learning models for multi-horizon forecasting using high-frequency sensor and weather observations, including temperature, humidity, rainfall/irrigation, solar radiation, wind, soil temperature, and electrical conductivity.
- Analyze large heterogeneous datasets and develop sensor-fusion, feature-engineering, and spatiotemporal learning approaches for extracting reliable information from noisy observations across locations, time periods, and sensing conditions.
- Calibrate and evaluate Hydroball, TEROS 12, SoilC 14, TEROS 21, and 5TE environmental sensors under South Florida field conditions to support reliable sensing and data integration.
- Apply HPC and GPU-based resources to train, optimize, and evaluate computationally intensive machine-learning and deep-learning models using large-scale imagery, geospatial, sensor, and spatiotemporal datasets.
- Design integrated AI, sensing, and IoT systems for real-time monitoring, forecasting, automated data processing, and decision support; develop cost-effective workflows for large-scale drone-image processing and stitching.
- Led the Fertilizer4Life project by analyzing soil-fertility data to support nutrient-management strategies and agricultural productivity.

Debre Markos University, Institute of Technology - Assistant Professor of Computer Science | *Ethiopia*, Jan 2021 - Jun 2024

- Led data-science and AI research across computer vision, predictive modeling, climate and environmental analysis, healthcare, and digital systems, with emphasis on transferable computational methods and real-world interdisciplinary applications.
- Served as Co-PI and project coordinator for the AI for Climate Action Innovation Research Network, applying AI/data-driven methods to weather, climate, and environmental datasets and coordinating interdisciplinary and international collaboration.
- Led or contributed to nationally and internationally funded projects, including the Smart Climate Project (Co-PI), AI Modeling for Maize Crop Disease Detection (PI), Digital Transformation Project (PI), and AI for Health Informatics Project (PI).
- Secured more than $100,000 in competitive research funding and coordinated multidisciplinary research teams and activities.
- Developed and supervised AI-based research in computer vision, crop disease detection, chronic-disease determinant analysis, cervical-spine disease classification, and data-driven decision-support systems.
- Taught undergraduate and postgraduate courses in Artificial Intelligence, Machine Learning, and Data Science, and supervised undergraduate, master's, and Ph.D. students.

- Served as Director of International Partnerships (2021-2024), establishing and coordinating research collaborations with institutions in the United States, Europe, and Asia.

National Yang-Ming Chiao Tung University, Department of Electrical and Computer Engineering - Postdoctoral Researcher | *Taiwan*, Feb 2020 - Dec 2020

- Developed a multimodal data-fusion system integrating spatial, temporal, visual, and mobility observations for real-time pedestrian detection and forecasting to support population safety.
- Conducted field experiments involving IoT signal measurements and drone-based image acquisition.
- Developed a system integrating energy-efficient drone base stations with ground network users for dynamic resource allocation.

National Taipei University of Technology, Department of Electrical Engineering and Computer Science - Research Scholar | *Taiwan*, Sep 2016 - Jan 2020

- Developed deep-learning, hybrid, and end-to-end data-driven models for localization, mobility prediction, object detection, and tracking in IoT and public-safety applications.
- Conducted field experiments involving UAV imagery acquisition and wireless-signal measurements across urban environments.
- Published four peer-reviewed journal articles during doctoral research.

Debre Markos University, Department of Information Technology - Lecturer and Researcher | Ethiopia | Sep 2015 - Aug 2016

- Taught undergraduate courses in Artificial Intelligence, C++, Python, C#, Networking, and Database Systems.
- Initiated and established a faculty research group on Artificial Intelligence and Smart Systems.

PEER-REVIEWED JOURNAL PUBLICATIONS

Full publication records are available through Google Scholar. Author names and bibliographic information are retained from the current CV; DOI/links are included where available.

1. **Adege, A. B***, *Abate,A.W., Nigussie, Y.M., Limenih, H.H.,* Eneyew,L.M., and Aycheh,D.A., (**2026**), Machine Learning-Based Sorghum Yield Prediction Using Multisource Data in Challenging Agricultural Environments of Ethiopia, Accepted on Oct 7, 2026. Discover Sustainability in Springer Nature.
2. **Adege, A. B***, Thangavel Murugan, Afedi Abdi Mohamed, N. Nasurudeen Ahamed, Zemzem Mohammed Megersa, Gadisa Alemu, Jemal Yusuf Hasen, and Nigusu Ademe Abebe, (**2026**). YOLO-Based Transfer Learning for Accurate and Efficient Pest-and-Disease Classifier of Maize Crop. https://doi.org/10.1016/j.array.2026.101170
3. Thangavel Murugan, Nasurudeen Ahamed N., Nura Shifa Musa, Eiman Masoud Alahbabi, Ruqayyah Ali Ahmed Alyammahi, **Adege, A. B**, Afedi Abdi, and Zemzem Mohammed Megersa, (**2026**). Research Advances in Maize Crop Disease Detection using Machine Learning and Deep Learning Techniques. Computers. https://doi.org/10.3390/computers15020099
4. Zemene Fikirte, **Adege, A. B**, Tewodros Getinet, Chimdo Anchala, and Getahun Wassie, (**2026**). Digital transformation maturity in Ethiopia's priority sectors. Discover Global Society, 4(1), 29. https://doi.org/10.1007/s44282-026-00347-0
5. Kebede, A. S., Muluneh, T. W., & **Adege, A. B*.** (2025). Detection of Weeds in Teff Crops Using Deep Learning and UAV Imagery for Precision Herbicide Application. Scientific Reports. https://doi.org/10.1038/s41598-025-15380-3
6. **Adege, A. B*** (2025). Advancing Biomedical Engineering with Artificial Intelligence and Machine Learning: Review. International Journal of Clinical Practice. https://doi.org/10.1155/ijcp/9888902

7. **Adege, A. B***, Admas, M. F., Tehone, Y., Tulu, B. B., & Endalamaw, E. T. (2025). AI-Powered Human Activity Detection and Tracking in Dense Crowds Using YOLOv8-DeepSORT. IET Image Processing, 19(1), e70227. https://doi.org/10.1049/ipr2.70227
8. Yimenu, T. K., **Adege, A. B***, & Techan, S. Y. (2025). Integrating Expert Knowledge with Machine Learning for AI-Based Stroke Identification and Treatment Systems. Digital Health, 11. https://doi.org/10.1177/20552076251336853
9. Limenih, H. H., **Adege, A. B***, Alene, A. Y., Demasu, H. T., & Belachew, H. M. (2025). Automatic Identification of Amharic Text Idiomatic Expressions Using a Deep Learning Approach. IEEE Access, 1. https://doi.org/10.1109/access.2025.3567811
10. Viriato, V., Rodrigues, G. S., Nunes, M. R., **Adege, A. B***, & Bonfim, F. P. G. (2025). On-farm observations of socio-environmental impacts of Humulus lupulus L. cultivation in Brazil. Agronomy Journal, 117(5), e70175. https://doi.org/10.1002/agj2.70175
11. **Adege, A. B*.** (2024). Real-Time Pedestrian Detection and Tracking in Disaster Areas Using Deep Learning Algorithms. Advanced Journal of Interdisciplinary Studies, 1206-1218. http://www.ajids.com.et/uploads/DMUJIDS-F-V7I23.pdf
12. Megersa, Z. M., **Adege, A. B***, & Rashid, F. (2024). Real-Time Common Rust Maize Leaf Disease Severity Identification and Pesticide Dose Recommendation Using Deep Neural Network. Knowledge, 4(4), 615-634. https://doi.org/10.3390/knowledge4040032
13. **Adege, A. B***, Lin, H., & Wang, L. (2020). Mobility Predictions for IoT Devices Using a Gated Recurrent Unit Network. IEEE Internet of Things Journal, 7(1), 505-517. https://doi.org/10.1109/jiot.2019.2948075
14. **Adege, A. B.**, Lin, H., Tarekegn, G. B., & Jeng, S. (2018). Applying Deep Neural Network (DNN) for Robust Indoor Localization in a Multi-Building Environment. Applied Sciences, 8(7), 1062. https://doi.org/10.3390/app8071062
15. **Adege, A. B***, Lin, H., Tarekegn, G. B., Munaye, Y. Y., & Yen, L. (2018). Indoor and Outdoor Positioning Using a Hybrid of Support Vector Machine and Deep Neural Network Algorithms. Journal of Sensors, 2018, 1-12. https://doi.org/10.1155/2018/1253752
16. Tarekegn, G. B., Juang, R., Lin, H., **Adege, A. B.**, & Munaye, Y. Y. (2020). DFOPS: Deep-Learning-Based Fingerprinting Outdoor Positioning Scheme in Hybrid Networks. IEEE Internet of Things Journal, 8(5), 3717-3729. https://doi.org/10.1109/jiot.2020.3024845
17. Yen, L., **Adege, A. B.**, Lin, H., Ho, C., & Lever, K. (2019). Deep Learning Approach to Channel Selection Strategy for Minimizing Co-Channel Interference in Unlicensed Channels. Microelectronics Reliability, 105, 113558. https://doi.org/10.1016/j.microrel.2019.113558
18. Munaye, Y. Y., Lin, H., **Adege, A. B.**, & Tarekegn, G. B. (2019). UAV Positioning for Throughput Maximization Using Deep Learning Approaches. Sensors, 19(12), 2775. https://doi.org/10.3390/s19122775

CONFERENCE PROCEEDINGS & PRESENTATIONS

1. **Adege, A. B.**, Awoke, G., & Bayabil, H. K. (2026, June 10-12). AI-Based Soil Moisture Forecasting and Field Capacity Detection Using Adaptive Feature Engineering for Precision Agriculture. ASABE 2026.
2. **Adege, A. B.**, Tulu, B. B., Teshome, F., Awoke, G., Hussain, S., & Bayabil, H. K. (2025). Streamlining Drone Imagery Analysis for Agricultural Decision-Making. Presented at the ASABE Annual Meeting, Jupiter, FL, June 11-13, 2025.
3. Boaz B. Tulu, Kinde A. Fante, Fathulak A. Shewajo, Eyosiyas T. Endalamaw, **Adege A. B.** (2025). Spatio-Temporal Deep Learning for Video Filtering and Safe Content Governance in Digital Platforms. EAI ICAST 2025 Proceedings. https://icast-conf.eai-conferences.org/2025/organizing-committee/
4. Eyosiyas T. Endalamaw, Kris Calpotura, Boaz Tulu, **Adege, A. B.**, Yared B. Legese (2025). CNN-Based Ethiopian Cuisine Classification: A Transfer-Learning Approach for Real-Time Mobile Deployment. EAI ICAST 2025 Proceedings. https://icast-conf.eai-conferences.org/2025/organizing-committee/

5. **Adege, A. B.**, Li, Y. R., Shuai, H. H., Lin, H. P., & Wang, L. C. (2020). Improving UAV Personalized-Tracking Services by Fusing Visual and Radio Data. IEEE ICPAI 2020. https://doi.org/10.1109/ICPAI.2020.00009
6. **Adege, A. B.**, Lin, H. P., Tarekegn, G. B., & Yayeh, Y. (2020). Mobility Prediction in Wireless Networks Using Deep Learning Algorithms. EAI ICAST 2019. https://doi.org/10.1007/978-3-030-55174-6_40
7. Tarekegn, G. B., Juang, R. T., Lin, H. P., Munaye, Y. Y., & **Adege, A. B.** (2020). Reducing Fingerprint Construction for Positioning IoT Devices Using Generative Adversarial Nets. IEEE ICPAI 2020. https://doi.org/10.1109/ICPAI.2020.00023
8. Munaye, Y. Y., **Adege, A. B.**, Tarekegn, G. B., Li, Y. R., Lin, H. P., & Jeng, S. S. (2019). Deep Learning-Based Throughput Estimation for UAV-Assisted Network. IEEE VTC 2019-Fall. https://doi.org/10.1109/VTCFall.2019.8891283
9. **Adege, A. B.**, Yen, L., Lin, H. P., Yayeh, Y., Li, Y. R., Jeng, S. S., & Berie, G. (2018). Applying Deep Neural Network (FFNN) for Large-Scale Indoor Localization. IEEE ICASI 2018. https://doi.org/10.1109/ICASI.2018.8394517
10. **Adege, A. B.**, Yayeh, Y., Berie, G., Lin, H. P., Yen, L., & Li, Y. R. (2018). Indoor Localization Using K-Nearest Neighbor and ANN Back Propagation Algorithms. IEEE WOCC 2018. https://doi.org/10.1109/WOCC.2018.8372899
11. Li, Y. R., Lin, H. P., Wang, L. C., **Adege, A. B.**, & Chen, L. Y. (2018). Multi-UAV Communication for Payload Data Link Enhancement Under Asynchronous Access. IEEE WOCC 2018. https://doi.org/10.1109/WOCC.2018.8372897
12. Yayeh, Y., Lin, H. P., Berie, G., **Adege, A. B.**, Yen, L., & Jeng, S. S. (2018). Mobility Prediction in Mobile Ad-Hoc Networks Using Deep Learning. IEEE ICASI 2018. https://doi.org/10.1109/ICASI.2018.8394506

MANUSCRIPTS UNDER REVIEW / IN PROGRESS

1. **Abebe B. Adege**, Girma W. Awoke, Boaz B. Tulu, and Haimanote K. Bayabil. "Soil Moisture Forecasting Using Ensemble and Deep Learning Models with Horizon-Adaptive Feature Engineering" (under review, minor).
2. **Abebe B. Adege** and Demeke Aschale Aycheh. "Multi-Satellite-Based Geospatial AI for Near-Real-Time Flood and Landslide Monitoring Using Deep Reinforcement Learning" (in progress).
3. **Abebe B. Adege**, Demeke A. Aycheh, Lijalem M. Eniyew, Haimanot K. Bayabil. "An End-to-End Fine-Tuned CNN-BiLSTM Hybrid Model for Early Detection and Classification of Apple Leaf Diseases" (under review).
4. **Abebe B. Adege**, Mengist, W. T., Bayabil, H. K., & Tulu, B. B. "An Ensemble Deep Learning-Based Framework for Intent Classification and Decision Support in Teff Farming" (under review).
5. Tulu, B. B., **Abebe B. Adege**, Endalamaw, E. T., & Lulu, J. B. "AI-Powered Real-Time Detection of Arabica Coffee Plant Diseases with Text and Voice-Based Recommendations in Ethiopia" (under review).
6. Feyisa Abebe Gelashe and **Abebe B. Adege**. "Machine Learning-Based Prediction of Under-Five Mortality Using Health, Socio-Demographic, and Climate Data in Eastern Hararghe, Ethiopia" (under review).
7. Muna Masresha Ayelgn and **Abebe B. Adege**. "A Cervical Spine Disease Severity Detection and Classification Using a Squeezed Convolutional Neural Network" (submitted for the Proceedings of the 14th EAI International Conference on Advancements of Science and Technology - EAI ICAST 2026).
8. Tirfie Birhane Woldaregay and **Abebe B. Adege**. "Enhancement of the Low Energy Adaptive Clustering Hierarchy (LEACH) Protocol to Increase the Lifetime of Wireless Sensor Networks" (in progress; submitted for the Proceedings of the 14th EAI International Conference on Advancements of Science and Technology - EAI ICAST 2026).

SELECTED RESEARCH GRANTS - FUNDED

Jun 2023 - Jun 2024 | AI for Climate Action Innovation Research Network. Funded by the West African Science Service Center on Climate Change and Adapted Land Use (WASCAL), Regional Universities Forum for Capacity Building in Agriculture (RUFORUM), and AKADEMIYA2063. $60,000. Co-PI and Project Coordinator. https://wordpress.ai4d.ai/project/ai-for-climate-action-innovation-research/

Sep 2023 - Aug 2024 | AI for African Maize Disease Detection and Classification. Funded by the United Arab Emirates University (UAEU). AED 20,000. Principal Investigator.

Jan 2023 - Jun 2024 | Digital Transformation Maturity in Ethiopia's Priority Sectors. Funded by the Ethiopian Policy Study Institute (PSI). ETB 2,000,000. Principal Investigator. https://psi.org.et/index.php/blog/305-assessing-a-digital-transformation-maturity-levels-the-case-of-priority-sectors-in-ethiopia

Jan 2022 - Dec 2024 | Digital Maturity Assessment of Ethiopia's Satellite, Spectrum, and Telecommunications Sector. Funded by the Ethiopian Policy Study Institute (PSI). ETB 1,800,000. Researcher. https://psi.org.et/index.php/blog/176-psi-released-its-preliminary-research-findings-on-the-maturity-of-internet-services-in-ethiopia

Dec 2022 - Jun 2024 | Digitalization of Indigenous Herbal Medicine in the Abay Gorge Region. Funded by Debre Markos University (DMU). ETB 10,000,000. Data Manager (Lead, Data Acquisition).

Mar 2019 - Jan 2020 | Deep Learning-Based Mobile User Tracking and Trajectory System via UAV. Funded by the Ministry of Science and Technology, Taiwan.

SELECTED RESEARCH SOFTWARE & AI SYSTEMS

- **N-DIP** - Nursery Disease Intelligence Platform: Developed a multi-disease early-warning platform integrating sensor and imagery datasets for real-world detection, monitoring, and phenotype-based diagnosis.
- **AgriSense AI** - Intelligent Sensing Application: Built an IoT-based desktop application integrating real-time sensor data, weather APIs, and machine-learning models for automated environmental estimation, monitoring, and alerts.
- **Deep Learning Studio:** Developed a desktop-based machine-learning research platform with a GUI-supported pipeline for CNN, RNN, and transformer model development from data preprocessing through deployment.
- **Coffee and Corn Disease Detection Systems:** Deployed mobile and desktop computer-vision applications for multiclass image-based disease detection using transfer learning, achieving more than 95% accuracy and providing field-oriented recommendations.

SELECTED LEADERSHIP & SERVICE

Debre Markos University - Director, International Partnership Directorate | *Ethiopia,* 2021 – 2024.

- Directed institutional partnerships with universities and research centers in the United States, Europe, and Asia.
- Led university exchange, joint research projects, international grant development, and capacity-building initiatives focused on research infrastructure and global engagement.

Debre Markos University - Coordinator, AI & Cross-Disciplinary Research Teams | *Ethiopia,* 2021 – 2024.

- Led multi-departmental research clusters applying AI in agriculture, health, climate adaptation, and related interdisciplinary domains.
- Promoted collaboration among faculty from computer science, agronomy, environmental science, and biomedical engineering.

Debre Markos University - Strategic Planning & Policy Committees | *Ethiopia*, 2020 - 2024

- Contributed to the university's five-year strategic plan for research and innovation.

- Helped draft institutional research policy on interdisciplinary collaboration and international engagement.

ACADEMIC JOURNAL LEADERSHIP & PEER REVIEW

- Associate Editor, Advanced Journal of Interdisciplinary Studies (2022-2024).
- Editorial Board Member, National Academic Journals, Ministry of Education - Ethiopia (2023-2024).
- Peer Reviewer for journals including IEEE Access, Digital Health, MDPI, Springer, and Elsevier titles.
- Conference Chair/Reviewer: ICT4DA (Track Chair, Wireless Networking), IEEE ICPAI, ICAST, and IEEE GLOBECOM.

HONORS & AWARDS

- International Graduate Student Scholarship supporting research in AI, IoT technology, and precision object detection in Taiwan; awarded by the Taiwan Government, 2016-2020.
- Best in University-Level Services Award for creating international partnerships, International Partnership Directorate, 2021-2024.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Certificate of Excellence for Partnership and Fund Securing, Debre Markos University of Technology, 2024.
- Ph.D. Equivalency Certificate, National Association of Credential Evaluation Services (NACES), USA, 2024.
- Preparing Future Faculty Program, University of Florida, Fall 2024.
- IEEE Elite School Certificate in Deep Learning and Computer Vision, Taiwan, 2018.
- Grant Proposal Writing and Research Development Training, Debre Markos University, 2021.
- Editorial Process and Journal Management Training, Debre Markos University, 2021.
- Reviewer and Contributor Recognition, ICT4DA Conference, Bahir Dar University, 2024.

TECHNICAL SKILLS

- Data Science & Machine Learning: Supervised and unsupervised learning, ensemble learning, statistical modeling, feature engineering, hyperparameter optimization, predictive analytics, model validation, time-series forecasting, spatiotemporal modeling, and heterogeneous data analysis.
- Remote Sensing & Geospatial Analytics: Satellite imagery, UAV imagery, Google Earth Engine, QGIS, ArcGIS, Pix4D, geospatial data integration, spatial analysis, mapping, and remote-sensing-based monitoring.
- Computer Vision & Deep Learning: CNN, RNN, GRU, BiLSTM, Transformer, YOLO, object detection, image classification, image segmentation, object tracking, feature extraction, transfer learning, and multimodal image analysis.
- Intelligent Sensing & Sensor Data Fusion: Environmental sensor networks, IoT integration, real-time data acquisition, weather and soil sensor integration, wireless measurements, sensor configuration, telemetry, heterogeneous sensor modeling, and multimodal fusion.
- High-Performance Computing & Large-Scale AI: HPC, GPU acceleration (CUDA), distributed computing, parallel data processing, large-scale model training, cloud-based workflows, computational optimization, and MLflow.
- Programming & Scientific Computing: Python, R, MATLAB, SQL, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras, statistical data analysis, and data visualization.

PROFESSIONAL LINKS

- Google Scholar: https://scholar.google.com/citations?user=OZc5w3YAAAAJ&hl=en
- LinkedIn: https://www.linkedin.com/in/abebe-belay-adege-9730701a0/